Major Front-End Migration for GRNsight, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

A gene regulatory network (GRN) consists of genes, transcription factors, and the regulatory connections between them which govern the level of expression of mRNA and protein from genes. A protein-protein physical interaction network (PPI) represents a set of proteins that bind to each other. A specialized software tool would help biologists better view and analyze such networks, so we created GRNsight, an open-source web application for visualizing models of GRNs and PPIs, either uploaded by a user or obtained from our backend budding yeast database. Graph models display rectangular nodes, representing genes/proteins, with edges connecting them. GRNs have directed red edges with pointed end markers or blue edges with blunt end markers to indicate regulatory relationships (activation or repression, respectively), with edge thickness indicating regulation strength. PPIs have undirected black edges to indicate physical binding. Time-course expression data can be displayed on GRN nodes to facilitate understanding of the regulatory relationships that generated that pattern of expression, creating new insights. GRNsight has been in development for over a decade, during which time both biological datasets and web technologies have advanced significantly. While the original implementation successfully supports the core functionality, it relies on older tools that make long-term maintenance and the addition of new features increasingly difficult. To address this challenge, GRNsight's JavaScript codebase has been migrated from a jQuery/Bootstrap/Pug framework to a modern React-based front-end which improves code readability, state management, and allows for more advanced interactive features. By modernizing the underlying framework, GRNsight is now positioned to continue expanding its functionality as new visualization needs arise for researchers to gain deeper insights into complex molecular systems. GRNsight React is available at https://dondi.github.io/GRNsight/react-thesis-4081.